Tutorial 1 — OLS Mechanics and Interpretation

Introductory Econometrics — Télécom Paris

Theory exercises

Exercise 1 (Deriving OLS)

Consider the simple regression model $y = \beta_0 + \beta_1 x + u$ and a sample $\{(x_i, y_i)\}_{i=1}^n$.

(a) Write the sum of squared residuals $S(b_0, b_1)$ and derive the two first-order conditions.

(b) Solve for $\hat{\beta}_0$ and $\hat{\beta}_1$. Which condition on the sample is needed for $\hat{\beta}_1$ to exist?

(c) Show that the first-order conditions are the sample analogues of the population moment conditions $\mathbb{E}(u) = 0$ and $\mathbb{E}(xu) = 0$.

(d) Deduce that $\sum_i \hat{u}_i = 0$, $\sum_i x_i \hat{u}_i = 0$, and that $(\bar{x}, \bar{y})$ lies on the fitted line. Are these facts evidence in favor of the model's assumptions?

Exercise 2 (Unbiasedness and failure of exogeneity)

Under assumptions SLR.1–SLR.4, write

$$\hat{\beta}_1 = \beta_1 + \frac{\sum_i (x_i - \bar{x}) u_i}{\sum_i (x_i - \bar{x})^2}.$$

(a) Prove this decomposition.

(b) Conclude that $\mathbb{E}(\hat{\beta}_1 \mid x_1, \ldots, x_n) = \beta_1$. State precisely which assumption makes the numerator have conditional mean zero.

(c) Derive $\mathbb{E}(\hat{\beta}_0 \mid X) = \beta_0$.

(d) Suppose instead that $\mathbb{E}(u_i \mid X) = \rho x_i$. Show that $\mathbb{E}(\hat{\beta}_1 \mid X) = \beta_1 + \rho$. If $\mathbb{E}(x_i) = 0$, verify that $\mathrm{Cov}(x_i, u_i) = \rho \mathrm{Var}(x_i)$.

(e) Suppose ability raises wages and is positively correlated with education, but ability is omitted from the wage regression. Use the decomposition to argue informally in which direction $\hat{\beta}_1$ is biased.

Exercise 3 (Unbiasedness and precision)

Let $Y_1, \ldots, Y_{10}$ be a random sample from a population with mean $\mu = 2$ and variance $\sigma^2 = 1$. Consider $\tilde{\mu}_1 = Y_1$ and $\tilde{\mu}_2 = \bar{Y}$.

(a) Show that both estimators are unbiased.

(b) Compute their variances. Which estimator is more precise, and by what factor?

(c) Generate 20 samples of size 10 from $N(2, 1)$. For each sample, record Y_1 and $\bar{Y}$; compare their means, ranges, and sampling distributions.

(d) Explain why an unbiased estimator can produce an estimate far from μ in one sample, and why comparing only average estimates hides an important difference between these two procedures.

Exercise 4 (Estimating the error variance)

Maintain SLR.1–SLR.5 and let $S_{xx} = \sum_i (x_i - \bar{x})^2$ and $SSR = \sum_i \hat{u}_i^2$.

(a) Starting from $\hat{u}_i = u_i - (\hat{\beta}_0 - \beta_0) - (\hat{\beta}_1 - \beta_1)x_i$, show that

$$SSR = \sum_i u_i^2 - n\bar{u}^2 - \frac{[\sum_i (x_i - \bar{x})u_i]^2}{S_{xx}}.$$

(b) Prove that $\mathbb{E}(SSR \mid X) = (n-2)\sigma^2$.

(c) Show that $\tilde{\sigma}^2 = SSR/n$ is biased and find its bias.

(d) Show that $\hat{\sigma}^2 = SSR/(n-2)$ is unbiased. Why are two degrees of freedom lost?

Exercise 5 (Log-level and level-log specifications)

Assume all variables inside logarithms are strictly positive.

(a) Let y change from y_0 to $y_1 = y_0 + \Delta y$. Show that

$$\Delta \log y = \log(y_1) - \log(y_0) = \log\left(1 + \frac{\Delta y}{y_0}\right).$$

Use a first-order Taylor approximation to explain why $\Delta \log y \approx \Delta y / y_0$ for a small relative change.

(b) Consider the log-level model $\log(y) = \beta_0 + \beta_1 x + u$. For a change Δx, derive the approximate percentage change in y and the exact percentage change between exponentiated fitted log values.

(c) Consider the level-log model $y = \beta_0 + \beta_1 \log(x) + u$. If x rises by $p\%$, derive the exact change in fitted y and its small-change approximation.

(d) In a log-level model let $\hat{\beta}_1 = 0.08$. Calculate the approximate and exact percentage changes for $\Delta x = 1$ and $\Delta x = 4$. In a level-log model let $\hat{\beta}_1 = 2.5$; calculate the approximate and exact changes in fitted y when x rises by 10%.

Python lab

Use `code/lab1.py` for all Python work. Extract the complete Tutorial 1 folder and keep its `code/` and `data/` folders together. The script locates `data/wage1.csv` automatically and creates `code/output/` for figures and tables. The guided blanks are marked `TODO`.

Lab 1. Seeing exogeneity and bias by simulation

Keep the same simulated values of x and repeatedly estimate the slope in $y = 1 + 0.5x + u$ under

$$\text{A: } u = \varepsilon, \qquad \text{B: } u = 0.4x + \varepsilon, \qquad \varepsilon \sim N(0, 4).$$

(a) Plot the two sampling distributions. Compare their mean slopes with 0.5 and with the values predicted by Exercise 2.

(b) Explain why one estimate can be far from 0.5 under A, why more replications do not make the distribution under B centre on 0.5, and which exogeneity condition holds under A but fails under B.

Lab 2. From data to a regression interpretation

Question. How do hourly wages vary with education in the observed sample, and how does the answer depend on the specification?

Data. The data contain 526 workers and 24 variables, based on the 1976 Current Population Survey. `wage` is hourly earnings in 1976 US dollars; `educ` is years of education. Every fitted model below is a simple regression with education as its only explanatory variable.

Step 1. Inspect before estimating

(a) Run `head()`, `shape`, `isna().sum()`, and `describe()` on wage and education. State the observation unit, sample size, wage units, and range of education.

(b) Compare mean and median wages. Check that wages are strictly positive before taking logarithms. Which observation count will every model use?

(c) Write exactly two descriptive sentences. In the first, identify what one observation represents, the sample size, and the source and year of the data. In the second, state the units of wage and education.

Step 2. Visualise and form a prediction

(a) Inspect histograms of wage and log wage and a wage–education scatterplot. What changes when the vertical variable is logged?

(b) Compute the wage–education correlation and predict the sign of the simple-regression slope. Does the graph suggest that education predicts every individual wage accurately? Write one sentence explaining the limitation.

Step 3. Fit and read a simple regression

Estimate `smf.ols("wage ~ educ", data=df).fit()`.

(a) Record the intercept and education slope rounded to three decimal places, n as an integer, and R^2 rounded to three decimal places. Then write the fitted equation using hats to distinguish estimates from parameters. Use `params`, `nobs`, and `rsquared`; formal tests come later.

(b) Interpret one additional year and four additional years of education in dollars per hour. Does this simple regression hold experience fixed?

(c) The intercept is negative. Must all fitted wages therefore be negative? Discuss the usefulness of the fitted value at zero education.

(d) Compute the slope using the centred-sums formula from Exercise 1. Check it against the software and add the fitted line to the scatterplot.

(e) Write exactly two interpretation sentences. The first must report the slope as dollars per hour for one additional year of education. The second must state that education is the only included regressor, so no other worker characteristic is held fixed and the result is an association, not a causal effect.

Step 4. Fitted values and residuals

(a) Calculate fitted wages at 12 and 16 years of education. Interpret their difference; is either value the known wage of every such worker?

(b) For the first observation calculate $\hat{u}_i = wage_i - \widehat{wage}_i$. Explain its sign. Why can we calculate this residual but not the population error?

(c) Check numerically that residuals sum to zero and that the fitted line passes through $(\overline{educ}, \overline{wage})$. Does this validate exogeneity?

(d) Inspect residuals against fitted values. Describe any visible change in spread without treating it as a formal test or as proof of a causal problem.

(e) Write one sentence that reports a fitted-value calculation and clearly distinguishes that fitted value from an individual outcome.

Step 5. Compare specifications and units

Estimate on the same sample:

$$\begin{aligned} \text{A: } & wage_i = a_0 + a_1 educ_i + v_i, \\ \text{B: } & \log(wage_i) = b_0 + b_1 educ_i + e_i, \\ \text{C: } & wage_cents_i = c_0 + c_1 educ_i + w_i, \qquad wage_cents_i = 100 wage_i. \end{aligned}$$

(a) Make a table with each model's outcome, slope, slope units, n, and R^2. Explain why slopes from A and B cannot be compared as dollar amounts.

(b) For Model B, use Exercise 5 to calculate both $100\hat{b}_1$ and $100\{\exp(\hat{b}_1)-1\}$ for one additional year of education. State clearly that the second number is the exact percentage difference between exponentiated fitted log wages. Then explain what extra restriction is needed before applying the same percentage to the conditional arithmetic mean $\mathbb{E}(wage \mid educ)$.

(c) Predict B's fitted log wages at 12 and 16 years and exponentiate them. Label the results "exponentiated fitted log wages." Why should we not automatically call them estimates of mean wages?

(d) B has a different R^2 from A. Can the larger value establish that B predicts wage in dollars better? Explain why the two denominators differ.

(e) Before estimating C, predict what happens to both coefficients, residuals, and R^2. Check your prediction. Does expressing wages in cents provide new economic information?

(f) Write three sentences: report the dollar association from A; report the approximate and exact percentage comparisons from B; and explain why neither estimate is automatically causal, naming one plausible omitted factor.